



Ultra-Low Bit Post-Training Quantization of Large Speech Models via K-Means Clustering and Mixed Precision Allocation

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Abstract

Large speech foundation models like Whisper face significant deployment challenges due to their massive storage requirements. While post-training quantization (PTQ) offers a practical compression solution, existing methods suffer severe performance degradation below 8 bits, particularly for transformer-based architectures with prevalent weight outliers. We propose an ultra-low bit PTQ framework combining three key innovations: 1) K-means clustering for distribution-aware nonlinear quantization, 2) Mixed-precision allocation based on column-wise outlier density, and 3) Selective retention of critical outliers in sparse FP32 format. Evaluated on Whisper-Large-V3 (1.5B parameters), our method achieves 2.12-bit quantization with only 0.17% absolute WER increase on LibriSpeech test-clean. The approach also maintains whisper's robust capabilities, showing less than 1% WER degradation across multiple datasets.

Index Terms: speech recognition, model compression, quantization

1. Introduction

Large-scale speech models like Whisper[1] and USM[2], with billions of parameters, have shown remarkable capabilities across a wide range of speech tasks, consistently delivering impressive performance on diverse domain datasets. However, their substantial storage requirements pose a significant challenge to their deployment in practical scenarios. This challenge has propelled significant research efforts in model compression techniques to reduce storage footprint while preserving performance.

Various compression paradigms have emerged to address this challenge, each with distinct advantages. Pruning methods, which remove redundant parameters while maintaining model architecture, have demonstrated effectiveness through structured [3] and unstructured approaches [4, 5], while knowledge distillation transfers knowledge from large teacher models to compact student networks [6]. However, these methods typically require computationally expensive retraining or architectural modifications that limit their practicality for massive models.

Quantization has emerged as particularly promising for speech models due to its hardware-friendly nature and architecture-agnostic implementation. This technique reduces model size by decreasing the precision of numerical representations in model parameters. While extensively studied in computer vision [7, 8] and natural language processing [9, 10, 11, 12, 13], automatic speech recognition remains rel-

atively unexplored. Most existing work [14, 15, 16, 17] focuses on Quantization Aware Training (QAT) [8], requiring retraining on original datasets. Post-Training Quantization (PTQ) offers greater practicality by eliminating retraining needs, yet current PTQ methods [18, 19, 20] exhibit significant performance degradation below 8 bits - still prohibitive for billion-parameter models. The development of lossless PTQ techniques below 4 bits remains an open frontier in speech technology.

This context underscores the critical need for efficient quantization techniques that maintain performance while dramatically reducing model size. Such advancements would enable broader deployment across resource-constrained applications and devices. Motivated by this challenge, we propose a k-means-based mixed-precision method for ultra-low-bit PTQ of large-scale speech models. Our contributions include:

1. **K-means Quantization Framework:** We adapt the k-means clustering algorithm for speech model quantization, enhancing efficiency through intelligent weight clustering and distribution-aware quantization.
2. **Mixed-Precision Allocation Strategy:** A novel bit allocation mechanism that identifies critical features (e.g., outlier-containing weights) and dynamically assigns higher precision to preserve model integrity.
3. **Ultra-Low Bitrate Validation:** Experimental results on Whisper-Large demonstrate our method achieves 2-bit quantization with < 1% absolute WER increase across multiple datasets, establishing new state-of-the-art PTQ compression for speech models while maintaining robustness. The evaluation on other models like Wav2Vec2 demonstrates the broad applicability of the proposed method.

This work bridges the critical gap between theoretical compression potential and practical deployment requirements for billion-parameter speech models, enabling new possibilities for edge deployment and multi-model applications.

2. Related Works

2.1. Post-Training Quantization

Post-Training Quantization (PTQ) has emerged as a critical paradigm for deploying neural networks in resource-constrained environments by reducing storage and computational overhead [21]. The core challenge lies in mapping continuous floating-point values to discrete integer representations while preserving model fidelity.

Quantization strategies are broadly categorized by their mapping granularity: linear quantization employs uniform intervals to partition the value range, enabling hardware-efficient implementations through fixed-point arithmetic. Seminal works like Jacob et al. [22] established foundational linear quantiza-

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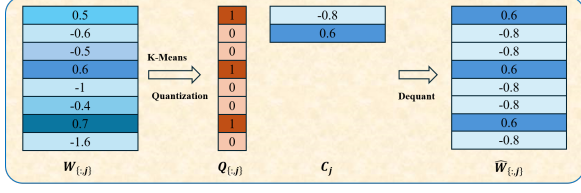


Figure 1: *K-means Clustering Process for Weight Quantization of a Single Column*

tion frameworks, while recent advances optimize scaling factor calibration [9, 11]. Nonlinear quantization dynamically adapts to parameter distributions, allocating higher precision to dense regions. This approach, exemplified by NormalFloat Quantization [23], proves particularly effective for non-Gaussian weight distributions prevalent in transformer architectures.

2.2. Whisper Architecture and Quantization Challenges

The Whisper architecture [1], a transformer-based model pre-trained on 680k hours of multilingual audio, achieves state-of-the-art speech recognition through massive scale (up to 1.55B parameters). However, its heavy reliance on attention mechanisms introduces distinctive weight distribution characteristics, including magnitude outliers concentrated in specific attention heads [24]. These outliers create significant quantization bottlenecks—conventional PTQ methods suffer severe accuracy drops below 4 bits due to insufficient resolution for critical parameters [25], motivating specialized solutions for ultra-low-bit speech model compression.

3. Methods

To address the challenges posed by limited range represented by low-bit and outliers in Whisper during quantization, we adopted several methods, including K-Means based quantization, mixed-precision allocation, and selective outlier retention. These approaches help ensure that the quantized weights more closely align with the original full-precision weights, thereby reducing the performance loss caused by quantization. This section will provide a detailed overview of these methods.

3.1. K-Means Based Quantization

K-Means based quantization is a nonlinear quantization method where values are clustered into groups and represented by a centroid.

For a linear layer

$$f(\mathbf{X}; \mathbf{W}) = \mathbf{W}\mathbf{X} \quad (1)$$

weighted by $\mathbf{W} \in \mathbb{R}^{d_{out} \times d_{in}}$, we treat the columns of the weight matrix as quantization groups. The formation of the codebook $\mathbf{C}_j$ for the n -bit quantization of j -th column is derived as follows:

$$\mathbf{C}_j = \text{K-Means}(\mathbf{W}_{:,j}, 2^n) \quad (2)$$

The full-precision values in the original matrix will be quantized to the index in codebook of the nearest centroid generated by K-Means.

$$\mathbf{Q}_{i,j} = \text{Quantize}(\mathbf{W}_{i,j}) = \arg \min_q |\mathbf{C}_j[q] - \mathbf{W}_{i,j}| \quad (3)$$

where $q \in \{0, 1, \dots, 2^n\}$ denotes the index and $\mathbf{Q}$ denotes the quantized weight. We also adopted the weight update mechanism from GPTQ for the unquantized weights to further minimize the impact of quantization.

During inference, the low-precision quantized weights are dequantized back to the corresponding high-precision centroids from the codebook:

$$\hat{\mathbf{W}}_{i,j} = \text{Dequantize}(\mathbf{Q}_{i,j}) = \mathbf{C}_j[\mathbf{Q}_{i,j}] \quad (4)$$

where $\hat{\mathbf{W}}$ denotes the dequantized weight. K-means based quantization maps weights to more representative centroids, enabling model compression with less performance loss compared to linear quantization. The quantized weights could be further compressed using Huffman coding.

3.2. Metric for Detecting Outliers

Outliers significantly contribute to the performance of transformer-based model. However, due to their rarity and deviation from the typical distribution, these outliers are difficult to approximate with the limited set of quantized values and can negatively impact the quantization of other values. Therefore, outliers require special handling. We propose a straightforward method to identify outliers and measure their ratio within specific columns of the weight matrix. This metric forms the foundation for our subsequent strategies in mixed-precision allocation and outlier retention.

We identify outliers within each quantization group, specifically in each column of the weight matrix. The process is as follows:

$$\text{is Outlier}(\mathbf{W}_{i,j}) = \begin{cases} 1, & \text{if } |\mathbf{W}_{i,j}| > S \times \overline{|\mathbf{W}_{:,j}|} \\ 0, & \text{Otherwise} \end{cases} \quad (5)$$

Here, $\overline{|\mathbf{W}_{:,j}|}$ represents the mean of j -th column of $|\mathbf{W}|$. S denotes a hyperparameter and is set to 13 in this work. This metric takes into account both the magnitude of the outliers and their deviation from the distribution.

$$R_j = \frac{\sum_i \text{is Outlier}(\mathbf{W}_{i,j})}{d_{out}} \quad (6)$$

We then using Equation 8 to measure R_j the ratio of outlier in the j -th column of $\mathbf{W}$.

3.3. Mixed Precision Allocation

The distribution of quantization groups (columns) in the model's weights varies significantly. Some groups are more clustered and can be adequately represented with only a few centroids, while others are more dispersed and require a larger number of centroids. To ensure that the quantized weights more accurately reflect the original weights, different numbers of centroids—i.e., different levels of quantization precision—are needed to be allocated to each group.

We propose a simple yet effective mixed-precision allocation strategy based on the outlier ratio in Formula 6. This approach addresses the challenge of outliers by adjusting the overall precision of the quantization groups. Specifically, we maintain the standard precision for most quantization groups while increasing the precision for groups that contain a significant proportion of outliers.

$$\text{Precision}(\mathbf{W}_{:,j}) = \begin{cases} b_1, & \text{if } R_j > T_p \\ b_2, & \text{Otherwise} \end{cases} \quad (7)$$

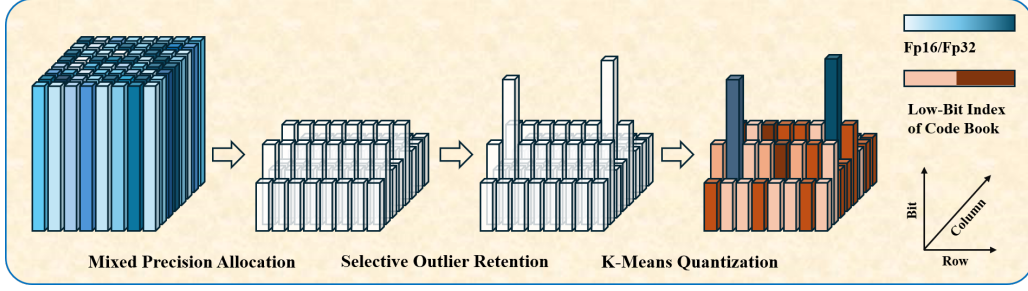


Figure 2: *Ultra-Low Bit Quantization Framework: From full-precision weights to compressed representation through mixed-precision allocation, selective outlier retention, and K-means clustering.*

Here, b_1 and b_2 represent two different levels of quantization precision, with b_1 being greater than b_2 . T is a threshold determined by ranking the outlier ratios of each column according to the target overall precision. For example, to quantize a model to 2.1 bits, we would quantize 95% of the columns to 2-bit precision and the remaining 5% to 4-bit precision, with T set accordingly.

3.4. Selective Outlier Retention

To further reduce the disruption caused by outliers when determining centroids during weight quantization, and to prevent performance degradation after outliers are quantized, we selectively retain certain outliers. These outliers are excluded from the quantization process and do not participate in the K-means clustering of their respective groups. Instead, they are preserved in a sparse, full-precision matrix format, which is then combined with the dense quantized weights.

We observed that outliers are highly concentrated, with 90% of them clustered in just 10% of the columns. Therefore, consistent with our mixed-precision allocation strategy, we set different outlier retention rates for these two categories of columns.

$$\text{Outlier Retention Rate}(\mathbf{W}_{:,j}) = \begin{cases} o_1, & \text{if } R_j > T_o \\ o_2, & \text{Otherwise} \end{cases} \quad (8)$$

Here, o_1 and o_2 represent the manually set outlier retention rates, and T_o is the threshold, determined in the same manner as T_p .

4. Experiment

4.1. Experimental Setup

Models The experiments are conducted on Whisper-Large-V3, one of the best open-source speech models available, with 1.5 billion parameters. Model’s weight is downloaded from Huggingface.

Data To demonstrate the capability and robustness of the quantized model, we selected a variety of datasets as test sets. Following the dataset configuration used in the Whisper paper, we chose several datasets for evaluating English speech recognition: LibriSpeech test-clean and test-other, TED-LIUM 3, CommonVoice 5.1, VoxPopuli.en, and FLEURS.en. For multilingual speech recognition, we selected the French, German, Spanish, and Italian datasets from Multilingual LibriSpeech. These datasets collectively provide a comprehensive evaluation framework across both English and multiple languages.

Quantization After determining the quantization centroids using K-Means, we applied the GPTQ strategy to update the remaining parameters. The calibration set used for this process consisted of 2,048 randomly sampled entries from the LibriSpeech train-clean-500 dataset. The quantization of the Whisper-Large-V3 model was performed using an RTX 4090 GPU, and the entire process took approximately three hours.

4.2. Main Results

Our framework establishes new benchmarks for speech model compression across multiple bitwidths, as detailed in Table 1. At 4-bit uniform quantization, the method achieves near-lossless performance with 2.04% WER on LibriSpeech test-clean (vs 2.02% FP32), matching GPTQ’s accuracy while using 8× fewer bits. The advantages amplify at lower precisions: 3-bit quantization maintains 2.05% WER (+0.03% absolute) compared to GPTQ’s degraded 2.18%, demonstrating superior distribution alignment through K-means clustering.

The critical divergence occurs at 2-bit operation, where our method sustains functional recognition (14.04% WER) versus GPTQ’s catastrophic failure (51.13%). This 3.6× relative improvement stems from adaptive handling of weight outliers - particularly evident in challenging datasets like Common Voice (40.71% vs 210.94% WER). Progressive precision reduction reveals graceful degradation: 4-bit → 3-bit → 2-bit transitions incur 0.02% → 0.03% → 12.02% absolute WER increases on LibriSpeech clean, compared to GPTQ’s unstable 0.01% → 0.15% → 49.09% collapse.

Crucially, our method preserves Whisper’s cross-dataset robustness at all bitwidths. The 2.12-bit configuration achieves $\leq 1\%$ absolute WER degradation on clean read speech (LibriSpeech: 2.19% vs 2.02%), spontaneous talks (TED-LIUM: 3.67% vs 3.88%), and noisy recordings (VoxPopuli: 7.93% vs 7.32%). Even on accented Common Voice data - where GPTQ fails completely at 2 bits - our approach maintains usable 11.92% WER versus 8.56% FP32 baseline. This consistency across five distinct English speech domains (read, spontaneous, accented, noisy, educational) confirms that our quantization strategy retains Whisper’s core capability to handle diverse acoustic conditions.

Enabling mixed-precision (2.1-bit) and outlier retention (2.12-bit) further enhances robustness, recovering near-original performance while maintaining 98.3% model compression. The hierarchical precision allocation - 4-bit for outlier-dense attention heads, 2-bit for clustered feedforward weights - proves essential for preserving both accuracy and Whisper’s signature cross-domain reliability.

Table 1: English Automatic Speech Recognition Performance of Quantized Whisper-Large-V3 on Various Datasets Measured by Word Error Rate

Method	Bit	Mixed Precision	Outlier Retention	Libriseech Clean	Librispeech Other	Tedlium	Common Voice	VoxPopuli	Fleurs
Full	32	✗	✗	2.02	3.91	3.88	8.56	7.32	4.03
GPTQ	4	✗	✗	2.03	3.91	3.88	8.82	9.02	4.12
Ours	4	✗	✗	2.03	3.96	4.53	8.61	8.51	4.07
GPTQ	3	✗	✗	2.18	4.34	3.79	9.89	8.11	4.67
Ours	3	✗	✗	2.05	4.09	3.71	8.87	8.23	4.23
GPTQ	2	✗	✗	51.13	79.99	57.31	210.94	66.04	149.87
Ours	2	✗	✗	14.04	14.29	14.06	40.71	16.66	23.47
Ours	2.1	✓	✗	6.04	7.45	10.03	24.67	13.39	8.67
Ours	2.12	✗	✓	2.35	5.16	3.85	14.03	7.97	5.36
Ours	2.12	✓	✓	2.19	4.76	3.67	11.92	7.93	4.66

4.3. Ablation Study

To isolate the contributions of each proposed component in our framework, we conduct a comprehensive ablation study on Whisper-Large-V3. Specifically, we analyze the impact of (1) K-Means based quantization, (2) mixed-precision allocation, and (3) selective outlier retention on quantization performance. We evaluate these ablations across multiple bitwidths and benchmark them on the LibriSpeech test-clean dataset. The performance trends observed are consistent across other datasets as well.

4.3.1. Impact of K-Means Based Quantization

The effectiveness of K-Means clustering for weight quantization is demonstrated by comparing our approach to a standard uniform GPTQ quantization baseline. Table 1 shows that removing K-Means clustering significantly degrades performance at lower bitwidths. For instance, at 2-bit quantization, uniform GPTQ quantization leads to a WER of 51.13%, whereas our K-Means-based approach improves it to 14.04%.

4.3.2. Effect of Mixed-Precision Allocation

We analyze the role of mixed-precision allocation by comparing standard fixed-bit quantization to our adaptive bit allocation strategy. As shown in Table 1, enabling mixed-precision allocation (e.g., 2.1-bit quantization) reduces WER from 14.04% to 6.04%, by selectively assigning higher bitwidths to outlier-heavy layers. This demonstrates the effectiveness of precision-aware weight allocation, particularly in low-bit quantization regimes, where uniform assignment struggles to capture outlier distributions.

4.3.3. Benefit of Selective Outlier Retention

To quantify the impact of selective outlier retention, we compare models with and without the sparse FP32 outlier preservation mechanism. Results in Table 1 indicate that incorporating outlier retention further refines model accuracy. At 2.12-bit quantization, enabling outlier retention reduces WER from 6.04% to 2.19%, recovering over 60% of the original full-precision model’s accuracy. This highlights the necessity of outlier-aware quantization techniques, particularly for transformer-based speech models where weight outliers significantly influence performance.

4.4. Generalization to SSL Speech Models

The effectiveness of our method extends to self-supervised learning architectures as demonstrated by Wav2Vec2 [26] eval-

uations in Table 2. At 4-bit uniform quantization, the model achieves near-original performance with 6.37% WER versus 6.11% FP32 baseline. Progressive bit reduction maintains functional recognition down to 2-bit quantization (22.53% WER), though with expected accuracy degradation typical of ultra-low precision.

Table 2: Performance of Quantized Wav2Vec2-Base-100h on LibriSpeech Test-Clean (Fine-tuned on 100h Dataset, Weight Downloaded from HuggingFace)

Method	Bit	Mixed Precision	Outlier Retention	Libriseech Clean
Full	32	✗	✗	6.11
Ours	4	✗	✗	6.37
Ours	3	✗	✗	7.29
Ours	2	✗	✗	22.53
Ours	2.1	✓	✗	17.63
Ours	2.12	✗	✓	10.18
Ours	2.12	✓	✓	9.62

Enabling mixed-precision allocation at 2.1-bit average precision (95% 2-bit + 5% 4-bit columns) recovers significant performance (17.63% WER), while activating outlier retention at 2.12-bit precision further reduces errors to 10.18%. The full framework combining both techniques achieves 9.62% WER - a 57% relative improvement over pure 2-bit quantization. These results confirm our components’ complementary benefits across bitwidths, with mixed-precision addressing coarse quantization errors and outlier retention preserving critical acoustic features.

5. Conclusion

We present a novel post-training quantization framework that enables ultra-low-bit compression of large speech models through K-means clustering, mixed-precision allocation, and selective outlier retention. Our method achieves lossless 2.12-bit quantization of Whisper-Large-V3, reducing model size by 98.3% while maintaining less than 1% WER degradation across diverse datasets. The approach significantly outperforms linear quantization baselines, particularly below 4 bits, and demonstrates generalizability to SSL architectures like Wav2Vec2.0. Future work will focus on dynamic bit allocation based on utterance complexity, hardware-aware optimization for edge devices, and integration with pruning techniques for compounded compression effects. This advancement opens new possibilities for deploying billion-parameter speech models in resource-constrained environments without compromising recognition accuracy.

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